RETRIEVAL LEVEL AND GENERATION LEVEL

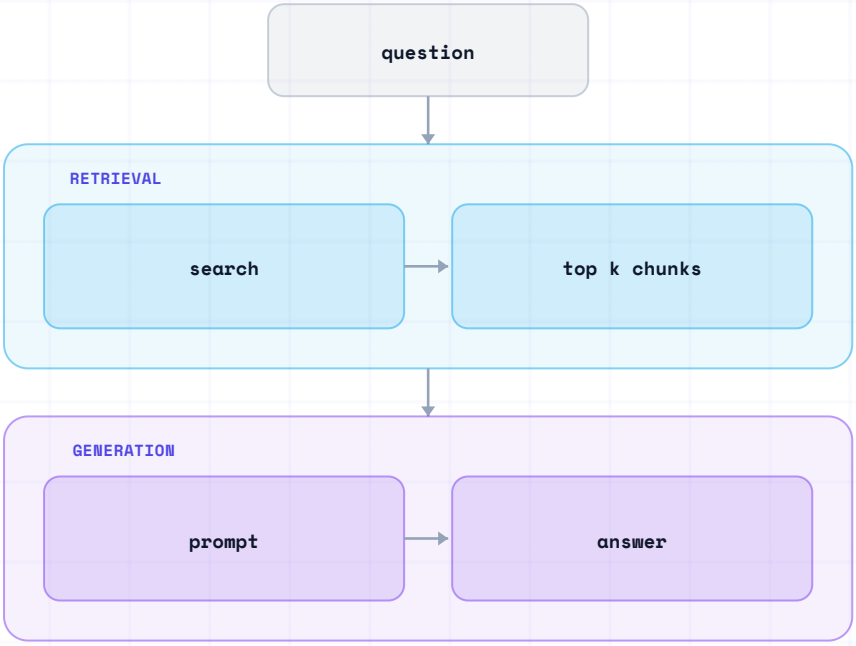
RAG Evaluation Metrics, Visually

Twelve metrics, drawn rather than defined. Which half they measure, what each number means, and which one to fix first.

PRACTICAI

Two Halves, Two Scorecards

SPLIT



two halves, two sets of metrics, measured apart

Retrieval: did the right text arrive

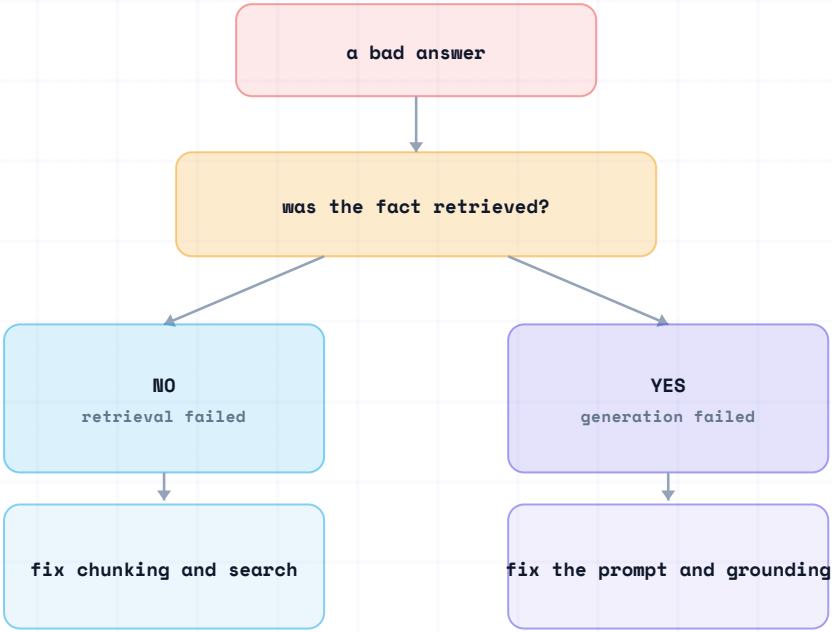
Generation: what was done with it

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Why Measure Them Apart

BLAME



one question splits every RAG bug in two

End to end scores tell you nothing about where

Split scores point at the component

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

The Six Retrieval Metrics

MAP

hit rate	did anything relevant appear
recall @ k	how much of the truth came back
precision @ k	how much of what came back was useful
MRR	how high was the first good one
NDCG	was the order right, with grades
context recall	did the context cover the answer

all six answer a different question about the same list

All computed from the same ranked list

And all need labelled expected chunks

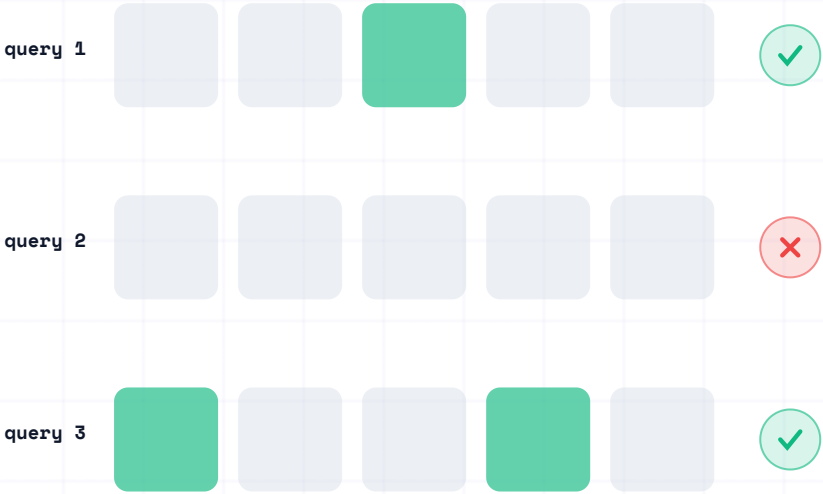
↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Hit Rate

DID ANYTHING LAND

DID ANY RELEVANT CHUNK MAKE THE TOP 5



hit rate = 2 of 3 queries

the crudest metric, and the first one to check

- Start here, it is the cheapest signal
- Says nothing about how many or how high

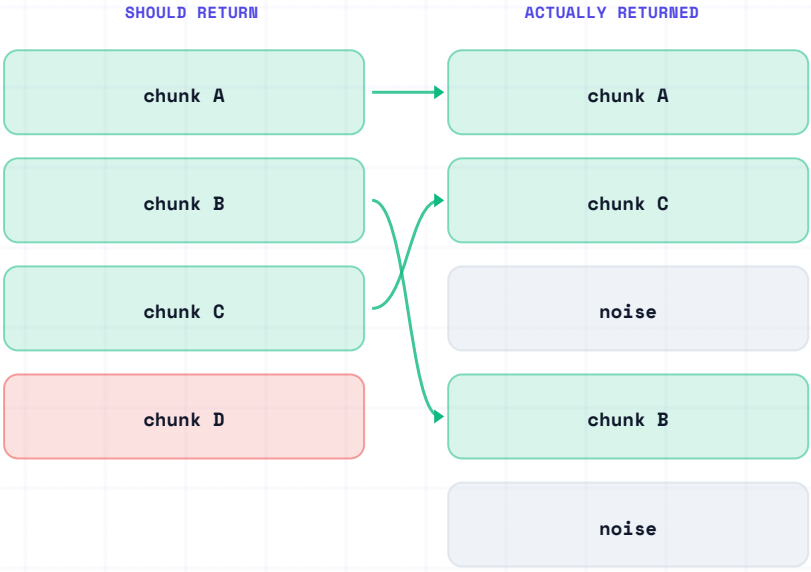
↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Recall @ k

COVERAGE

OF EVERYTHING THAT SHOULD COME BACK



recall @ 5 = 3 of 4 = 0.75

chunk D was never found, so no prompt can use it

The single most important retrieval number

A chunk never retrieved can never be used

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Precision @ k

NOISE

OF WHAT CAME BACK, HOW MUCH WAS USEFUL



precision @ 6 = 3 of 6 = 0.50

low precision means noise in the prompt and wasted tokens

Low precision fills the prompt with noise

And you pay for every noisy token

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

MRR

HOW HIGH

HOW FAR DOWN WAS THE FIRST GOOD ONE



MRR = mean of those three = 0.61

rewards putting the right chunk first, not just somewhere

- Rewards ranking the right chunk first
- Best when there is one correct answer

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

NDCG

GRADED ORDER

A HIT AT RANK 1 IS WORTH MORE THAN AT RANK 8



NDCG also lets a chunk be partly relevant, not just yes or no

the honest metric when relevance has degrees

- Handles partly relevant chunks
- The metric a reranker should move
- Normalised, so 1.0 is a perfect order
- Needs graded labels, not just yes or no

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

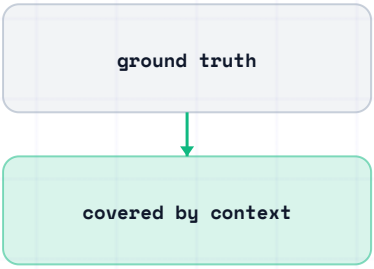
PRACTICAI

Context Precision and Recall

RAGAS

CONTEXT RECALL

did the context contain the answer

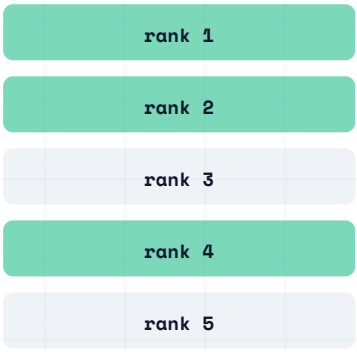


4 of 5 claims covered

= 0.80

CONTEXT PRECISION

were the useful chunks near the top



relevant ones ranked high

the two RAGAS metrics people confuse most often

- Both scored against the ground truth
- Both need a judge, not a label
- Recall is coverage, precision is order
- The names mislead almost everyone

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Which One to Watch

CHOOSE

IF YOU WANT TO KNOW	TRACK
Is retrieval working at all	Hit rate
Am I missing the answer	Recall @ k
Am I stuffing the prompt with noise	Precision @ k
Is the ranking any good	MRR or NDCG
Did a reranker help	NDCG, before and after
Does the context cover the answer	Context recall

AND THE ORDER TO FIX THEM

1 Recall first, always

2 Then precision

3 Then ranking

⊘ Never the reverse

> A perfect ranking of the wrong chunks scores well on MRR and answers nothing. Recall is the metric with a veto.

If you track one number, track recall at 5.

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

The Six Generation Metrics

MAP

faithfulness	is every claim in the context
answer relevance	does it answer the question asked
completeness	did it cover every part
correctness	does it match the known answer
citation accuracy	do the citations really support it
refusal accuracy	does it decline when it should

these need a judge, because there is no single right string

No single correct string exists

So a judge or a human has to score them

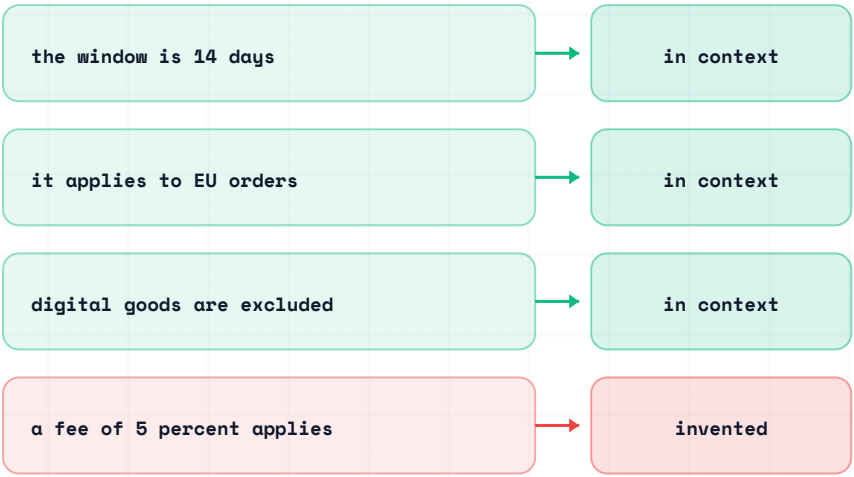
↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Faithfulness

GROUNDEDNESS

SPLIT THE ANSWER INTO CLAIMS, CHECK EACH ONE



faithfulness = 3 of 4 = 0.75

one unsupported claim is one hallucination

The metric that measures hallucination

Score claims, never whole answers

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Answer Relevance

ON TOPIC

TRUE, GROUNDED, AND STILL NOT AN ANSWER

what is the refund window?

14 days from delivery	✓	answers it
our refund policy is important to us	✗	true, but no
the shipping window is 3 days	✗	wrong question

faithfulness and relevance fail independently

- An answer can be true and useless
- ↔ Faithful and relevant are independent

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Completeness

COVERAGE

A MULTI PART QUESTION NEEDS A MULTI PART ANSWER

what is the window, the fee, and the process?

the window	the fee	the process
✓	✓	✗

completeness = 2 of 3 = 0.67

a partly answered question still scores full marks

on faithfulness and relevance

- Score part by part, not as a whole
- List the required parts when you label

> A half answer scores full marks on faithfulness and relevance. Only this metric sees the missing third.

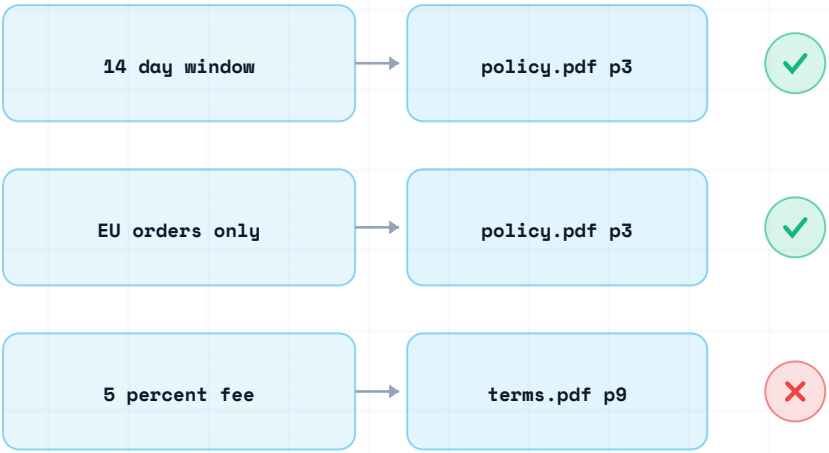
↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Citation Accuracy

PROVENANCE

DOES THE CITED PAGE ACTUALLY SAY IT



a citation that does not support the claim is worse

than no citation at all

Check that the cited page says it

A wrong citation buys false trust

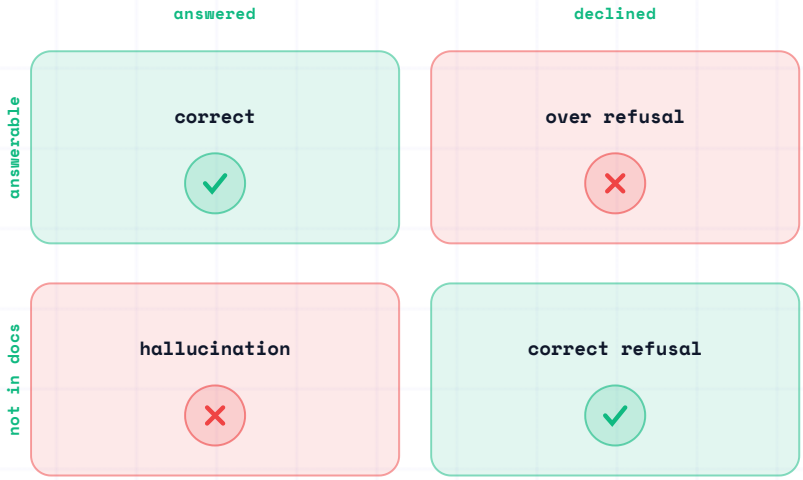
↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Refusal Accuracy

SAYING NO

THE ANSWER IS NOT IN THE DOCUMENTS



most eval sets contain no unanswerable questions at all

- Two failure modes, not one
- Add unanswerable questions to your set
- Hallucination is the expensive corner
- Over refusal is the annoying one

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Why BLEU and ROUGE Fail

WORD OVERLAP

SAME MEANING, DIFFERENT WORDS

reference

you have 14 days to return it
you have 14 days to return it

model answer

returns are accepted within two weeks

BLEU 0.08 ROUGE-L 0.21 a human says correct

an LLM judge says: faithful, relevant, correct

word overlap metrics punish correct paraphrase

They score wording, not meaning

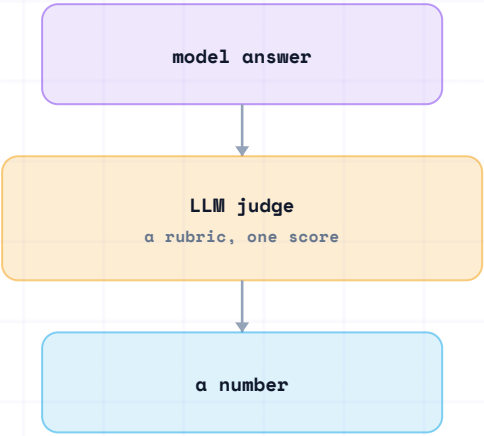
A judge costs money and is worth it

> If a metric punishes a correct paraphrase, it is measuring the wrong thing for RAG.

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

The Judge

LLM AS EVALUATOR



BUT FIRST, CALIBRATE IT

100 examples scored by a human, then by the judge

if they disagree, the judge is the thing that is broken

an uncalibrated judge is a random number generator

Calibrate against human labels first

Then use a cheap model to run it at scale

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

Building the Eval Set

FOUNDATION

WHAT GOES IN THE SET

- real production queries
- edge cases you already hit
- multi part questions
- unanswerable questions
- expected chunk ids
- expert written answers

50 to start, 200 to trust, 500 to catch regressions

labelled once, reused forever

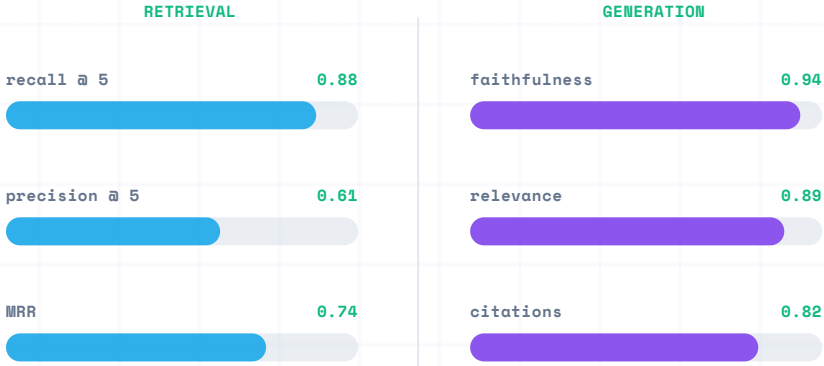
- Sample from real traffic, not your imagination
- Every incident becomes a new case

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

What to Put on the Dashboard

TRACKING



precision @ 5 is the weak one here

so the prompt is carrying noise it does not need

track both halves, on the same chart, over time

- Track both halves side by side
- The weakest bar is your next task

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT

PRACTICAI

The Metric Map

- Hit rate
- Recall @ k
- Precision @ k
- MRR
- NDCG
- Context recall
- Faithfulness
- Answer relevance
- Completeness
- Citation accuracy
- Refusal accuracy
- A calibrated judge

Anyone can say the answer was wrong.
Engineers can say
which half was wrong.

Save this before your next RAG evaluation.

↓ CHECK THE CAPTION FOR THE FULL INTERVIEW KIT